ExEnDiff: An Experiment-Guided Diffusion Model for Protein Conformational Ensemble Generation

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Understanding protein conformation is key to understanding their function. Importantly, most proteins adopt multiple conformations with nontrivial ensemble distributions that change depending on their environment to perform functions like catalysis, signaling, and transport. Recently, machine learning techniques, especially deep generative models, have been employed to develop protein conformation generators. These models, known as unified protein ensemble samplers, are trained on the Protein Data Bank (PDB) dataset and can generate diverse protein conformation ensembles given a protein sequence. However, their reliance solely on structural data from the PDB, which primarily captures folded protein states, restricts the diversity of the generated ensembles and can result in physically unrealistic conformations. In this paper, we overcome these challenges by introducing ExEnDiff, an experiment-guided diffusion model for protein conformation generation. ExEnDiff integrates experimental measurements as a physical prior, enabling the generation of protein conformations with desired properties. Our experiments on a variety of fast-folding and intrinsically disordered proteins demonstrate that ExEnDiff significantly advances the capabilities of current unified protein ensemble samplers. With little computational cost, ExEnDiff can capture important proteins' configuration properties and the underlying Boltzmann distribution, paving the way for a next-generation molecular dynamics engine. We further demonstrate the effectiveness of ExEnDiff to capture conformational changes in the presence of mutations and as an efficient tool for determining a reasonable collective variable space for protein ensembles. With these results, ExEnDiff is well poised to push the study of protein ensembles into a data-rich regime currently available to few problems in biology.

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I. INTRODUCTION

The dynamic nature of biological macromolecules is fundamental to their capacity to perform essential functions within the cell. For proteins in particular, these conformational changes are crucial because they regulate protein activity and dictate interactions with other biomolecules, allowing precise control in complex physiological processes such as signaling, transport, and gene expression. Understanding this conformational flexibility is therefore essential for comprehending how proteins function in both normal physiological states and pathological conditions, offering insights that are critical for advancing therapeutic development. However, our ability to capture this conformational variability is hindered by the limitations of current experimental and computational methods. Experimental techniques like nuclear magnetic resonance (NMR) and cryo-EM are resource intensive and often miss transient states, while computational approaches like molecular dynamics require significant computational power to explore relevant timescales.

Recent advancements in artificial intelligence (AI) have transformed structural biology, particularly within the problem space of determining a folded structure from amino acid sequence alone. AI-based approaches to this problem, such as AlphaFold [1,2], RoseTTAFold [3], trRosetta [4], and ESMFold [5], have significantly enhanced our ability to predict and analyze protein conformations with unprecedented accuracy. More recent efforts have extended beyond single-structure prediction to generate protein conformational ensembles. Initial methods, such as MSA subsampling [6] and clustering [7], have attempted to diversify AlphaFold's predicted conformations by adjusting model inputs. More advanced strategies have focused on deep generative models, particularly diffusion models [8,9], to address the challenge of generating protein conformation ensembles [10–13]. These models operate on the premise that metastable states of proteins are encoded within the native structures of other proteins. By training on large datasets of native protein structures from

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the Protein Data Bank (PDB), these models use a stochastic perturbation process to explore conformation space with greater diversity and accuracy. This emerging class of deep generative models, referred to as unified protein ensemble samplers, shows great promise as an avenue for generating protein ensembles [14].

However, PDB predominantly contains experimentally resolved, folded protein structures. This limits the ability of the database to capture the full spectrum of protein conformations, particularly in less-structured regions, Thus, unified protein ensemble samplers trained solely on PDB data struggle to accurately explore the entire conformational landscape of proteins. At the time of AlphaFold2, the PDB had ∼140 000 well-curated structures (and evolutionary couplings were trained on a database of more than two billion clustered protein sequences) providing a wealth of training data to learn static structures. For protein ensembles, this wealth of data is not available. Furthermore much of the experimental data available for ensemble properties of proteins are ensemble measurements that are difficult to connect to a molecular representation of the ensemble. Computational methods like molecular dynamics (MD) simulation can provide access to protein ensembles [15–18] but are computationally intensive, especially for large proteins [19]. Efforts to integrate physical knowledge, such as using MD energy as a prior [20,21], have been made to align sampled conformations with the underlying Boltzmann distribution. However, these approaches often require costly retraining and are constrained to implicit solvent force fields, limiting their accuracy and broader applicability.

In this study, we introduce an experiment-guided diffusion model for protein conformation generation (ExEnDiff), a deep generative unified protein ensemble sampler that leverages experimental measurements to guide the protein conformation sampling process. ExEnDiff seamlessly integrates experimental data without incurring additional computational overhead and is compatible with any state-of-the-art (SOTA) deep generative unified protein ensemble sampler. The core innovation of ExEnDiff lies in its ability to direct the sampling process in alignment with experimental measurements, ensuring that the generated ensembles closely adhere to desired physical properties and accurately reflect the underlying Boltzmann distribution. The key technique behind ExEnDiff is guided sampling within a diffusion model, a widely adopted approach in generating images, videos [22–25], and biomolecular structures [26–28] using diffusion models.

We demonstrate the efficacy of ExEnDiff through extensive protein conformation generation experiments on a range of fast-folding and intrinsically disordered proteins, achieving high-quality sampling without requiring model retraining. Numerical evaluations show that ExEnDiff consistently outperforms a SOTA unified protein ensemble sampler, especially for proteins with flexible and disordered regions. We propose that ExEnDiff serves as a versatile computational tool for rapidly generating protein ensembles that closely follow the Boltzmann distribution, with broad applications in tasks such as collective variable selection, force-field parametrization, and high-throughput screening.

II. RESULTS

A. Overview of the ExEnDiff model

The high-level methodology framework of ExEnDiff is illustrated in Fig. 1. Similarly to other unified protein ensemble samplers, ExEnDiff leverages a diffusion model, as depicted in Fig. 1(a). The diffusion model perturbs the data distribution toward a simple Gaussian distribution using a stochastic differential equation (SDE) and then learns the score function in the reverse SDE to recover the original data distribution. Unified protein ensemble samplers utilize the stochasticity of the diffusion model to generate diverse protein conformation ensembles through multiple steps of the reverse diffusion process. However, without guidance, these samplers may produce ensembles that significantly deviate from the true Boltzmann distribution. Inspired by restrained ensemble models [29–31] and experiment-guided MD simulations [32–35], ExEnDiff addresses this limitation by incorporating experimentally measured observables into the sampling process as a correction term, as shown in Fig. 1(b). During the reverse diffusion process, these experimental measurements guide the sampler to align closer to the true experimental values, as depicted in Fig. 1(c). As a result, ExEnDiff enables the unified protein ensemble sampler to capture both critical global and local features of protein ensembles, generating conformations that more accurately follow the Boltzmann distribution compared to guide-free samplers. Formulations of the diffusion model and ExEnDiff are detailed under Methods, with more details in Supplemental Material [36], Sec. 1.

In this work, we employ Str2str [12] as the pretrained unified protein ensemble sampler due to its high performance in generating feasible protein ensemble configuration. However, it is important to emphasize that ExEnDiff is not confined to a specific unified protein ensemble sampler; it is designed to be adaptable to any ensemble sampler that leverages a diffusion model. As advancements are made in pretrained unified protein ensemble samplers, ExEnDiff will likewise benefit, further enhancing its performance.

1. Experimental data improves the accuracy and diversity of generated protein conformations

To demonstrate the effectiveness of ExEnDiff, we assembled a comprehensive benchmark set of proteins, comprising two ordered proteins (BBA [37] and Villin [38]) and eight intrinsically disordered proteins (IDPs), including ACTR [39], drkN SH3 [40], NTail [41], Aβ40 [42], the ParE2-associated antitoxin PaaA2 [31], the intrinsically disordered region from the *Saccharomyces cerevisiae* transcription factor Ash1 [43], RS peptide, and ELF3 [44]. This diverse set of proteins contains a diverse range of secondary structures, including α helices and β strands, and various lengths from 24 to 234 amino acids. Furthermore, this benchmark set represents a broad spectrum of proteins, from ordered to highly disordered, and poses a significant challenge for current unified protein ensemble samplers. The ground-truth Boltzmann distribution was obtained from long, unbiased MD simulations, except for ELF3, where it was derived using Replica Exchange with Solute Tempering (REST) simulations to enhance sampling

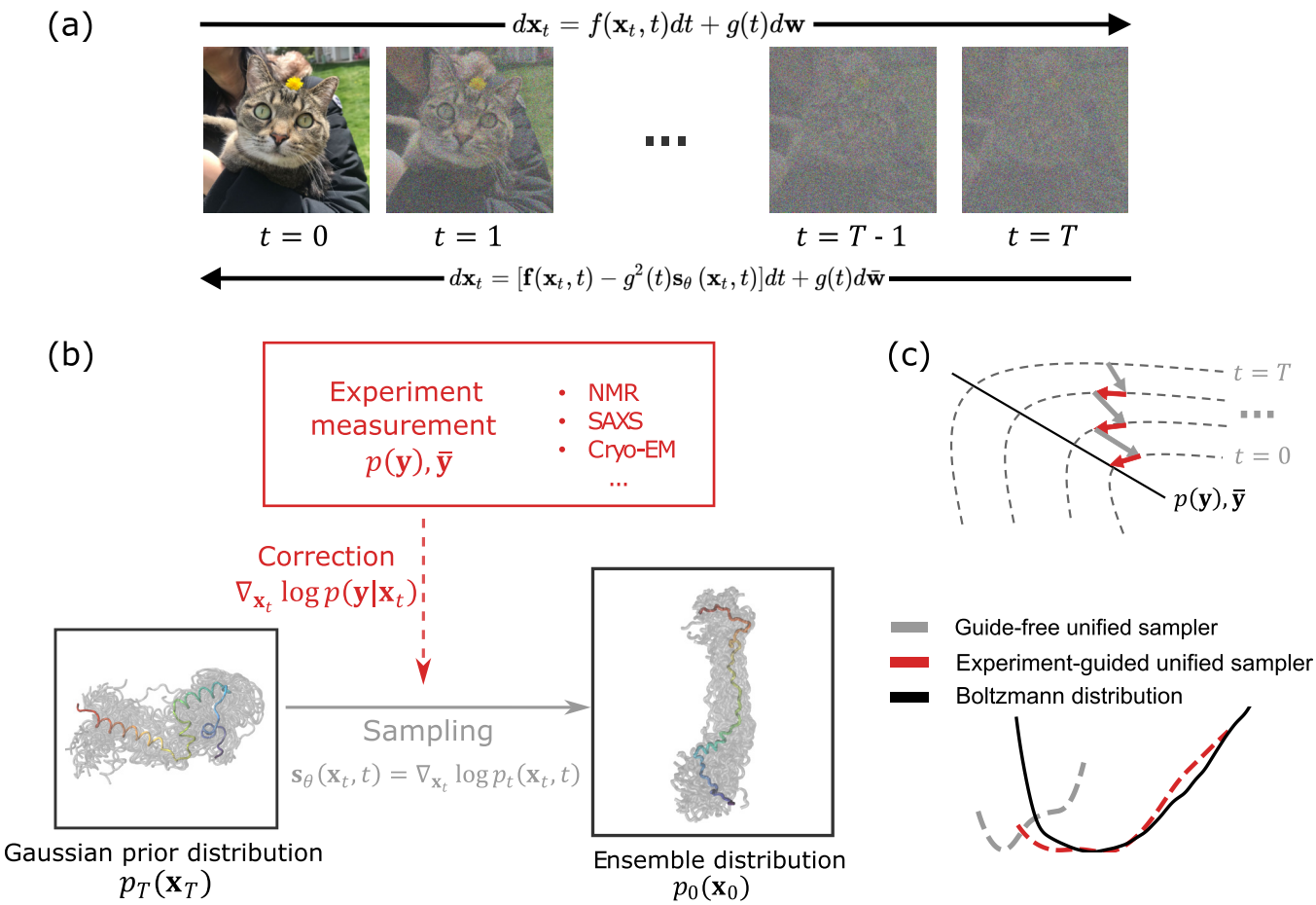

FIG. 1. The high level framework of ExEnDiff. (a) A visual demonstration of score-based diffusion model: Perturb the data distribution to a simple prior distribution through a forward stochastic differential equation and learn the reverse diffusion process to recover the correct data distribution. (b) Unified protein ensemble samplers generate initial samples $\mathbf{x}_T$ from a prior Gaussian distribution, and the reverse diffusion process gradually converts the prior configurations into ensemble configurations $\mathbf{x}_0$. ExEnDiff incorporates a correction term into the pretrained score function $\mathbf{s}_\theta(\mathbf{x}_t, t)$, from measurements $\mathbf{y}$ from various experimental techniques, including NMR, small-angle x-ray scattering (SAXS), and cryo-EM. (c) A guide-free unified protein ensemble sampler typically fails to sample protein conformation spaces that follow the Boltzmann distribution. In contrast, by guiding the sampling process of a pretrained unified protein ensemble sampler with experimental measurements, ExEnDiff generates configurations that get more closely align with measurement distribution $p(\mathbf{y})$ or measurement average $\bar{\mathbf{y}}$ through the reverse diffusion process. This enables ExEnDiff to effectively explore conformation spaces, producing protein conformation ensembles that are closer to the true Boltzmann distribution.

of diverse conformational states (simulation details of all proteins are included in the Supplemental Material [36], Sec. 2, Fig. S1 and Fig. S2). In this work we have also introduced a large RS-peptide trajectory (net sampling of 1.76 ms) for the research community, which was generated using the Folding@home distributed computing system [45,46].

We examine the ability of ExEnDiff to generate conformational ensembles that follow Boltzmann distributions by utilizing a wide range of experimentally measurable parameters as guiding metrics. Here we test the performance of ExEnDiff guided by three sets of guiding metrics: (1) distribution of end-to-end distance (EED), (2) distribution of radius of gyration (Rg), and (3) a combination of distribution of radius of gyration and the secondary structure, specifically the percentage of α helix per residue (Rg + SS). These parameters can be inferred from either a single or a combination of experiments, such as Förster resonance energy transfer (FRET) [47–53], small-angle x-ray scattering (SAXS) [54–58], small-angle neutron scattering [59–62], circular dichroism spectroscopy [63–67], and NMR experiments [68–70], for which further details have been provided in Supplemental Material [36], Sec. 3. For all proteins in the experiment, we generate 100 distinct protein ensemble conformation samples from the unified protein ensemble samplers. The conformation generation takes less than 300 s on a single A100 GPU in all experiments. We note that with additional computational resources, this number can be increased to capture more conformation ensembles.

To provide a clear and straightforward indicator of the performance of ExEnDiff guided by different experiments and the guide-free sampler, we present the accuracy using a single metric—sampler score, as shown in Fig. 2(a). This score is calculated based on the discrepancy between the observables produced by the samplers and those from ground-truth MD simulations. For a sampler, a lower sampler score indicates the generated protein conformational ensembles produce close

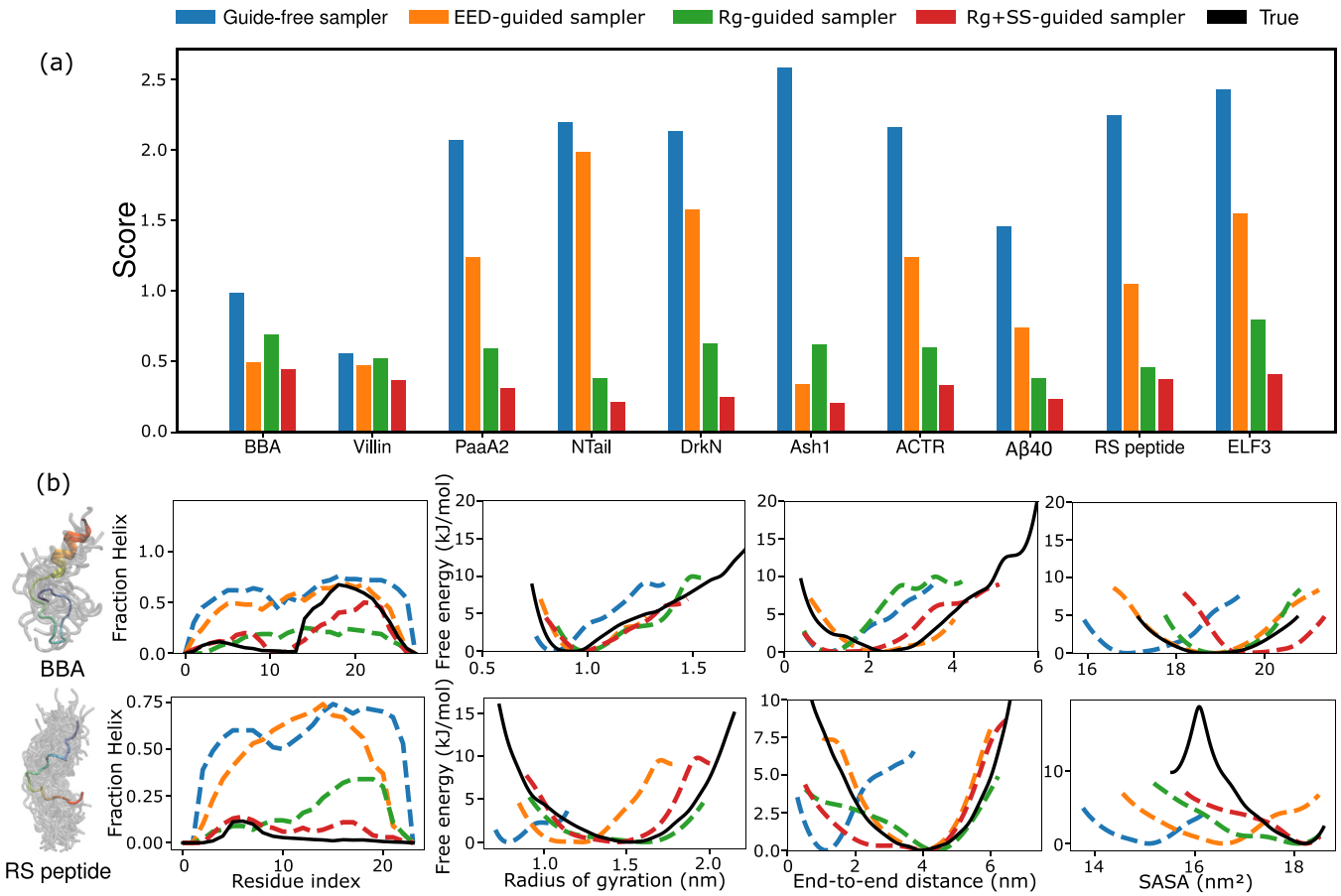

FIG. 2. Results for guide-free and different experiment-guided unified protein ensemble sampler EED (end-to-end distance), Rg (radius of gyration), and SS (secondary structure). (a) Unified protein ensemble sampler scores for folded and disordered proteins from the benchmark examined in this work. (b) Fraction helix, free energy surface of radius of gyration, solvent-accessible surface area, and end-to-end distance of two representative proteins from the benchmark: BBA and RS peptide.

agreement in observables from ground-truth MD simulations. The ground-truth Boltzmann distribution was obtained from long unbiased MD simulations. Simulations of Villin, ACTR, drkN SH3, NTail, $A\beta 40$, PaaA2, and Ash1 were conducted by DE Shaw research using the a99SB-disp force field [71] for 30 μs. The force-field parameters were optimized based on experimental measurements to achieve high accuracy in simulation results across various ordered and disordered proteins. Simulations of BBA and Villin were conducted by DE Shaw research [72,73] using the CHARMM22* force field and the modified TIP3P water model [74] compatible with the CHARMM [75] force field for 325 and 125 µs, respectively. For ELF3-PrD, the converged free-energy surface was inferred from the REST2 simulations [76], which is an unbiased enhanced simulation method. The details of the simulation protocol are listed in Lindsay *et al.* [44]. Finally, for the RS peptide, we ran long unbiased simulations of up to 2 ms of simulation time, with 20 unique starting structures, each run with 100 clones of ∼0.9 μs each using Amber03ws force field [77] and Tip4p/2005 water model [78]. The sampler score is calculated based on the discrepancies in end-to-end distance, radius of gyration, solvent-accessible surface area, root-mean-square fluctuation (RMSF), and helix percentage between the sampler and ground-truth MD simulations, as detailed under Methods and Supplemental Material [36], Sec. 4. The results show that ExEnDiff, guided by the experimentally measured parameters, significantly enhances sampling quality. This improvement is particularly notable for disordered proteins, where the original sampler struggles. Notably, ExEnDiff guided by both the radius of gyration and secondary structure (Rg+SS) achieves the best performance, as it integrates both global (radius of gyration) and local (secondary structure) information, greatly enhancing the model's capacity to explore relevant protein conformation space.

We further demonstrate critical physical features using BBA and RS peptide in Fig. 2(b) as representatives of their particularly class of protein: ordered and disordered, respectively (results for all proteins studied are available in Supplemental Material [36], Sec. 5 and Fig. S3–S6). While the guide-free sampler performs well in generating feasible ensembles for BBA, ExEnDiff improves the model by refining local secondary structures. Conversely, the guide-free sampler performs poorly on RS peptide, a highly disordered protein, where it tends to sample helixlike structures. This illustrates that a unified protein ensemble sampler trained on PDB data tends to generate foldlike conformations. With the inclusion of experimental measurements, ExEnDiff generates ensembles with more accurate physical features. Importantly, even guiding the sampler with a single parameter, such as the

TABLE I. Comparison of Jensen-Shannon (JS) distances with respect to the two-dimensional UMAP for each sampler and protein in the experiment. Red indicates the sampler with the lowest JS distances for each protein.

	BBA	Villin	PaaA2	NTail	DrkN	Ash1	ACTR	Aβ40	RSpeptide	ELF3
Guide-free	0.50	0.51	0.40	0.44	0.37	0.29	0.35	0.47	0.50	0.44
EED-guided	0.49	0.52	0.21	0.40	0.34	0.32	0.31	0.40	0.41	0.39
Rg-guided	0.40	0.46	0.38	0.38	0.30	0.30	0.24	0.27	0.41	0.31
Rg+SS-guided	0.36	0.33	0.18	0.24	0.27	0.27	0.29	0.33	0.26	0.41

radius of gyration, can significantly improve both global and local aspects of the generated protein ensembles.

Finally, we examine the capacity of ExEnDiff in capturing the complete conformational ensemble with relative weights. To assess this, we calculate a two-dimensional (2D) UMAP representation of each protein using ground-truth MD simulation data and project the ensembles generated by each sampler onto this space. The 2D UMAP effectively captures critical features of the conformational landscape, allowing for a direct comparison of how well each sampler explores relevant states. We demonstrate the Jensen-Shannon (JS) distances of the UMAP between the ground-truth MD simulation and each sampler in Table I. The table shows that by including the experimental measurement guidance, ExEnDiff consistently generates ensembles that better align with the MD simulation than the original Guide-free sampler. This highlights the advantage of integrating experimental constraints, leading to improved fidelity in conformation space exploration.

We note that all experimental measurements used to guide the protein ensemble generation process in this study are synthetically derived from ground-truth MD simulations. In real experiments, secondary-structure populations per residue can be accurately predicted from NMR chemical shifts using the δ2d program [79]. For end-to-end distance and radius of gyration, we provide a preliminary discussion on utilizing real experimental measurements when certain key features are missing in Supplemental Material [36], Sec. 6 and Fig. S7.

B. Capturing mutant protein conformations using ExEnDiff

Generating accurate protein conformational ensembles for protein mutants has long posed a challenge for deep learning models trained on the PDB. Small changes in protein sequences can lead to significant structural discrepancies, which deep learning models can struggle to capture effectively. In this section, we evaluate ExEnDiff's ability to generate structurally distinct protein conformational ensembles for a mutant of Chignolin [80]. This mutant replaces the original tyrosine residues at both termini with glycine and features a threonine-to-proline mutation at position eight, hereon it will be referred to as T8P. These mutations are known to destabilize the native hairpin conformation [81,82], providing a rigorous test of ExEnDiff's capability in handling structurally divergent mutants.

While the guide-free sampler accurately identifies the native structure of wild-type Chignolin and the local destabilization at the mutation site, it fails to account for the global structural changes induced by the mutation, producing similar conformations for the mutant, as shown in Fig. 3. In contrast, when the sampling process is guided by the radius of gyration, ExEnDiff successfully captures the global destabilization of the β hairpin in the T8P mutant, as reflected in the fraction of β structure. Guidance by radius of gyration alone, also improves the prediction of end-to-end distance and solvent-accessible surface area, providing a more accurate representation of the mutant's conformational ensemble.

1. Cryo-EM 2D images enhance the sampling of diverse and physically realistic conformations

As a final proof-of-concept of an experiment ExEnDiff can be guided by, we extend the capacity of our model to use 2D images from cryoelectron microscopy (cryo-EM) [83–88] as guiding parameters. Cryo-EM has become an important experimental technique for generating high-resolution 3D structures of biological molecules. In cryo-EM experiments, hundreds of thousands of noisy 2D images with random orientations are captured of a sample. Recent studies have suggested that cryo-EM has the potential to extract conformational ensembles that follow the equilibrium distribution of the sample at the temperature before freezing [26,89–94]. These methods either use reconstructed map to guide MD simulations, or combine 2D images and MD simulations as a prior for ensemble reweighting. Either process can become computationally prohibitive when dealing with large-scale proteins.

In this work, we use cryo-EM 2D images as guidance for the unified protein ensemble sampler. We adopt a simple model to represent the configuration-to-image formulation process in cryo-EM following previous work [92]:

$$I(\mathbf{x}) = \mathcal{P}_z \mathcal{R} \rho(\mathbf{x}), \tag{1}$$

where $\rho(\mathbf{x})$ is the electron density as a function of the protein configuration $\mathbf{x}$, $\mathcal{R}$ the rotation matrix of the electron density, and $\mathcal{P}_z$ the projection along z axis. For simplicity, the electron density $\rho(\mathbf{x})$ is modelled as the sum of spherically symmetric 3D normal densities centered on the $C\alpha$ of the protein.

We generate synthetic images corresponding to each configuration from the ground-truth MD simulations and add multiple levels of noise to form the cryo-EM dataset that guides the sampling process of unified protein ensemble samplers. In this experiment, we assess the performance of ExEnDiff when guided by cryo-EM images across three different signal-to-noise ratio (snr) levels: $[1, 0.1, \infty]$, where $\text{snr} = \infty$ represents a noise-free scenario. The results, illustrated in Fig. 4 with additional experiments available in the Supplemental Material [36],, reveal several key findings. While the guide-free sampler generates configurations with limited diversity, as evidenced by the radius of gyration, ExEnDiff demonstrates a significantly enhanced ability to explore the protein conformation space, particularly in capturing less compact conformations. Additionally, ExEnDiff shows a

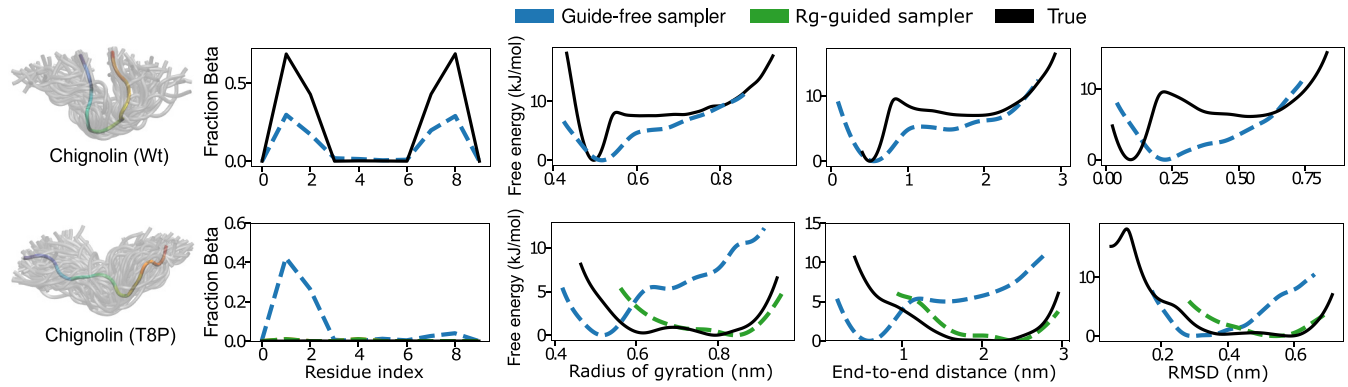

FIG. 3. Performance of ExEnDiff guided by radius-of-gyration in capturing mutants' effect, evaluated by fraction helix, free energy surface of radius of gyration, solvent-accessible surface area, and end-to-end distance using wild type Chignolin and its T8P mutant.

strong capacity for predicting correct secondary structure, as indicated by the fraction of helical content.

Remarkably, ExEnDiff produces similar results when guided by cryo-EM images with no noise (snr $= \infty$) and those with an snr of 1, indicating its robustness to noisy inputs. This suggests that ExEnDiff has substantial potential for application to real cryo-EM data, which typically contains significant levels of noise. We hypothesize that this resilience to noise arises from the nature of the reverse diffusion process, which inherently functions as a denoising mechanism, thereby enabling ExEnDiff to effectively mitigate noise in cryo-EM images during the protein conformation generation process.

However, we emphasize that experimental guidance using cryo-EM images is still in its initial proof-of-concept stage, and several limitations, further discussed in Supplemental Material [36], Sec. 7, must be addressed in future work before this method can be effectively applied to real noisy micrographs.

2. Efficient training of data-driven collective variables with ExEnDiff

In the final part of the experiment, we showcase a potential application of ExEnDiff, namely the development of efficient data-driven collective variables (CVs) for enhanced sampling simulations [95–106]. Compared to traditional physics-based CVs, data-driven CVs can more effectively capture critical low-dimensional features of complex systems. However, a significant challenge with ML-based CVs lies in the quality of the training data, which is often limited by insufficient exploration of the full conformation space of proteins. The capacity of ExEnDiff to generate protein conformational ensembles that reasonably cover protein conformation space makes it an appealing tool as the starting point for training data-driven collective variables for protein ensembles. Inspired by AlphaFold2-RAVE [107], we used ExEnDiff to generate 100 initial samples of a protein conformational ensemble, guided

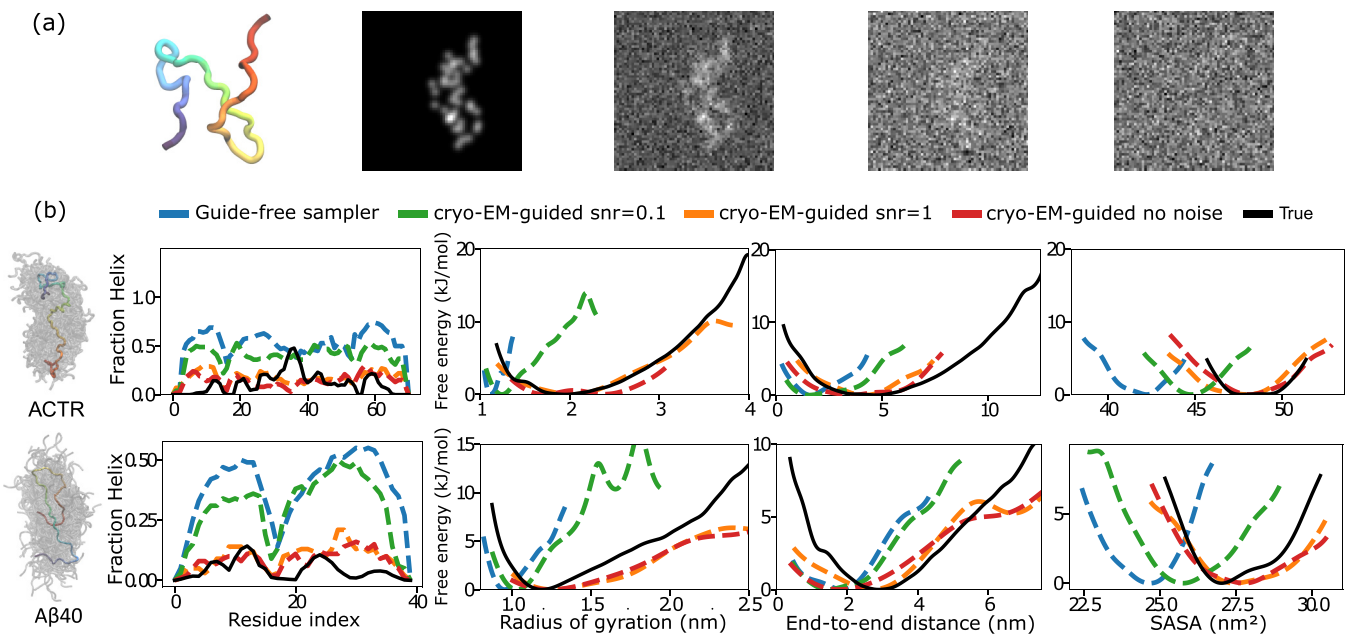

FIG. 4. (a) Visualization of synthetic cryo-EM 2D images of an Aβ40 configuration with different levels of signal-to-noise-ratio (snr): from left to right is snr $= \infty$ (no noise), snr $= 1$, snr $= 0.1$, snr $= 0.01$. (b) Performance of ExEnDiff guided by cryo-EM 2D density images of snr $= \infty$ (no noise), snr $= 1$, snr $= 0.1$, evaluated by fraction helix, free energy surface of radius of gyration, solvent-accessible surface area, and end-to-end distance of two representative proteins from the benchmark: ACTR and Aβ40.

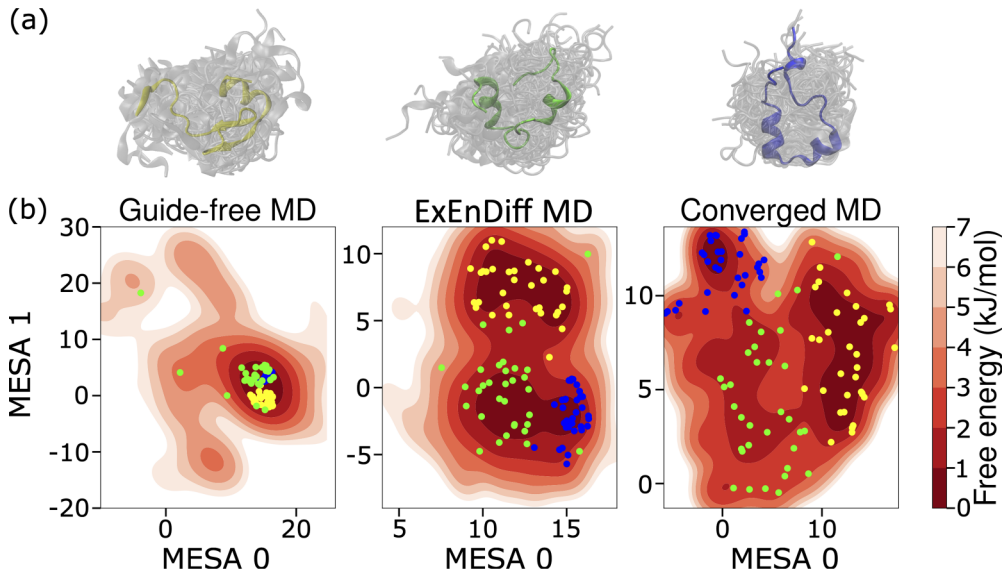

FIG. 5. (a) The three major conformational states of Villin, identified by the k-means clustering algorithm, are as follows: the yellow cluster predominantly features ensembles with one major helix at the start of the protein; the green cluster includes ensembles with two major helices, one at the start and one in the middle; and the blue cluster represents ensembles with three major helices located at the start, middle, and end of the protein. (b) Free energy surface of the 2D MESA collective variables (CVs) obtained from training on three independent MD trajectories for Villin. The datasets include the following: (1) one hundred unique starting configurations generated by the guide-free sampler, each run for 5 ns, resulting in a total of 500 ns of MD simulation (guide-free MD); (2) 100 unique starting configurations generated by the ExEnDiff sampler, guided by the radius of gyration and secondary structure, each run for 5 ns, resulting in 500 ns of MD simulation (ExEnDiff MD); (3) a 325-μs MD simulation by DE Shaw Research (Converged MD). The dots, in three different colors, represent the three major conformational states clustered as described in (a), and projected onto each free energy surface.

by the radius of gyration and secondary structure. Each sample was subsequently subjected to a short 5-ns MD simulation, resulting in a cumulative 500 ns of MD simulation data. These data were then used to train 2D data-driven CVs. For the CV training, we adopted MESA [108], one of the deep learning–based approaches that uses an autoencoder-based framework to extract low-dimensional features as CVs. Details of CV training can be found in Supplemental Material [36], Sec. 8.1 and Fig. S8. The same data collection process was applied to the guide-free protein ensemble sampler, and we used the same neural network architecture to train the CVs for both samplers and the converged MD.

One key indicator that a CV is effective is the clear separation of major states into distinct clusters, which reflects the capture of significant configurational similarities and differences. In this study, we applied the k-means clustering algorithm to identify three major conformational states from the converged ground-truth MD simulation (a 325-μs MD simulation by DE Shaw Research). These clusters represent distinct secondary structures, with specific helix patterns, as shown in Fig. 5(a). To assess the quality of the CVs, we projected the configurations from the ground-truth MD trajectory onto the 2D CV space of each sampler and plotted the corresponding free energy surface, as illustrated in Fig. 5(b). Compared to long unbiased MD simulations, which require microsecond timescales, ExEnDiff identifies multiple major states using significantly fewer computational resources. In contrast, the guide-free sampler captures only a single major conformational state, highlighting its limited ability to detect diverse states.

III. DISCUSSION AND CONCLUSIONS

In this work, we have introduced ExEnDiff, a unified protein ensemble sampler that can generate protein ensemble configurations aligning with correct Boltzmann distribution by incorporating experimentally inferred data. At its core, ExEnDiff's use of experimental guidance shares significant similarities with molecular dynamics simulations with experimental restraints (described in Supplemental Material [36], Sec. 9). In MD simulations, experimental restraints are often used to constrain or guide the conformational sampling through restraining certain variables (like distances, angles, or volumes) to ensure that the simulated ensemble remains consistent with experimental observations. This technique is especially valuable when certain aspects of the system's conformational space are not well sampled or when high-resolution data is limited. Experimental restraints in MD can take many forms—distance and angular restraints from chemical shifts, dipolar couplings, and Nuclear Overhauser Effect data; cryo-EM or x-ray crystallography derieved restraints aimed to ensure simulated structure fits the density maps and atomic positions; FRET, hydrogen-deuterium exchange, and SAXS to restrain distances and shapes [109–113]. Similarly explicit corrective potentials derived from secondary-structure assignments have been used to guide MD simulations [114]. While the experimental guides introduced in our method follow a similar philosophy, the key advantage is a more efficient exploration of conformational space by directly generating diverse ensembles based on experimental data, without running expensive full MD simulations.

We anticipate a broad range of future work building on the ExEnDiff framework. As reported under Results, ExEnDiff excels in generating physically meaningful protein conformational ensembles, thereby enhancing the exploration of protein conformation space. This makes ExEnDiff an ideal engine for generating training data for data-driven CVs.

Another significant research direction is the optimization of the force-field parametrization pipeline. Traditionally, force-field parametrization involves iteratively adjusting force-field parameters to align with experimental measurements, followed by rerunning MD simulations using the updated force field. This process is both time-consuming and computationally expensive. Recent advances in deep generative models have shown promise in parametrizing potential energy functions [115,116]. However, these methods are constrained by their reliance on data from converged MD simulations, limiting their ability to improve beyond the force field that generated the initial MD data. ExEnDiff offers a solution by generating protein conformation ensembles that align with various experimental measurements. By integrating ExEnDiff's ability to produce experimentally consistent and Boltzmann-weighted ensembles with traditional force-field components that maintain the underlying physical principles, along with deep generative models that accurately compute full-body potentials, we can achieve a more efficient and accurate approach to protein force-field parametrization. This combination could represent a substantial advancement in the field, significantly reducing the time and computational resources required while improving the quality of the resulting force fields.

ExEnDiff can be particularly useful in the data analysis pipelines for single particle cryo-EM, that in principal allows access to the underlying conformational ensemble of the sample making it invaluable for studying proteins and complexes that exist in multiple functional states. A long-standing problem in this direction has been to extract free energy profiles from cryo-EM particles [117]. More recently, an ensemble reweighting approach has been developed that estimates ensemble density from 2D cryo-EM particle data by reweighting based on prior molecular dynamics ensembles [92]. These approaches rely on a prior, generally from MD, to thoroughly sample the full conformational ensemble, which can be hard for larger proteins and complexes—the target systems in most cryo-EM experiments. Here ExEnDiff and ExEnDiff supplemented with short MD can provide an attractive alternative by presenting an informative prior through a more efficient exploration of conformational space.

Beyond cryo-EM, along the same vein, the ensembles generated from ExEnDiff can be further refined by integrating a broad range of direct experimental measurments through tools such as HDXer and BioEN [118–122].

In the current version, ExEnDiff focused on structural information obtained after processing raw experimental data, often through established physical and mathematical models. For example, a commonly used method to extract Rg from SAXS data is the slope from the Guinier plot at small scattering vector regimes [57,123,124]. However, this approach comes with several assumptions, such as the homogeneity and compactness of the sample. For complex or irregularly shaped particles (e.g., elongated, branched, or highly anisotropic structures), the Guinier approximation may not be applicable, leading to inaccuracies in the structural interpretation. Therefore, arguably, ExEnDiff with raw experimental data, rather than processed outputs, could be more informative at guiding towards relevant protein structures. Given that ExEnDiff is a versatile tool capable of handling diverse data types, as previously demonstrated, future iterations could take advantage of this flexibility by directly incorporating raw experimental data into the inference process. This would allow ExEnDiff to overcome the limitations of or supplement traditional model-based methods.

Despite its significantly improved performance, ExEnDiff has a few limitations. One is the use of a simple harmonic restraint potential in manifold constraint sampling, which ignores covariance between dimensions. A more advanced approach, such as Mahalanobis-distance-based restraints, could address this problem and will be explored in future iterations. Another limitation is that ExEnDiff relies on samples from the measurement distribution rather than the full distribution itself, which can be problematic for high-dimensional data such as the 2D density images obtained from cryo-EM experiments if the samples are not fully representative. To mitigate this, future efforts will focus on developing a density-guided diffusion model. In this vein, another promising future direction is to explore the model's limits with larger protein structures and extending its capability to handle polynucleotides and biologically relevant macromolecular assemblies. Finally, although ExEnDiff is capable of generating diverse and multimodal protein conformational ensembles, it does not directly produce metastable states with highly accurate free energy differences. This highlights the need for a better unified protein ensemble sampler. The presented protocol is flexible to incorporate any off-the-shelf diffusion model-based protein ensemble sampler. Moreover choosing better and/or unprocessed guiding experimental measurements, and advancing the effectiveness of guidance strategies as described above will be crucial for refining ExEnDiff to better capture the underlying free energy landscape.

Overall, we have demonstrated that ExEnDiff is an effective tool for accessing conformational distributions of protein ensembles using simple experimentally obtained values as a guide. We believe this approach will be key to pushing the study of protein ensembles into the data regime required for the large-scale learning of not just static structures but true protein ensembles. This will accelerate our ability to understand the effects of protein environment and mutations on protein ensemble properties, currently an outstanding problem in the field, with tremendous implications for both basic science and human health.

IV. MATERIALS AND METHODS

A. Score-based diffusion models for unified protein ensemble sampler

The current unified protein ensemble samplers [10–13] all use diffusion models as the base generative model. The score-based diffusion model perturbs data with original distribution $p_0(\mathbf{x})$ to Gaussian distribution $p_T(\mathbf{x})$ with a diffusion process

over a unit time horizon by a SDE:

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)dt + g(t)d\mathbf{w}, \quad t \in [0, T], \tag{2}$$

where $\mathbf{f}(\mathbf{x}_t, t)$, $g(t)$ are chosen diffusion and drift functions and $\mathbf{w}$ denotes a standard Wiener process. For any diffusion process in Eq. (2), it has a corresponding reverse-time SDE [125]:

$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g^2(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t, t)]dt + g(t)d\bar{\mathbf{w}}, \tag{3}$$

with $\bar{\mathbf{w}}$ a standard Wiener process in the reverse-time. The score-based diffusion model learns $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t, t)$ with a parameterized score function $\mathbf{s}_\theta(\mathbf{x}_t, t)$. To sample from the data distribution $p_0(\mathbf{x})$, we draw samples from the prior distribution $p_T(\mathbf{x}) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and then discretize and solve the reverse-time SDE with numerical methods, e.g., Euler-Maruyama discretization. Details of the score-based diffusion model used in this paper can be found in Supplemental Material [36], Sec. 1. In this study, we focus on the variance preserving formulation of the SDE used in the denoising diffusion probabilistic model (DDPM) [8]:

$$\mathrm{d}\mathbf{x} = -\tfrac{1}{2}\beta(t)\mathbf{x}dt + \sqrt{\beta(t)}\mathrm{d}\mathbf{w}, \tag{4}$$

where $\beta(t)$ denotes the variance schedule. When discretized, DDPM utilizes a predefined variance schedule $\beta_1, \beta_2, \ldots, \beta_T$. By defining $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$, the forward diffusion process can be expressed as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}). \tag{5}$$

The primary objective of all unified protein ensemble samplers is to learn and generate independent and identically distributed protein configuration samples from the underlying distribution $p(\mathbf{x}|\text{seq})$, which represents the conformational space associated with a given protein sequence. Achieving this objective is particularly challenging due to the inherent difficulty in obtaining a comprehensive set of proteins with conformation samples that accurately follow the Boltzmann distribution.

B. ExEnDiff: A manifold constraint sampling protocol for guiding unified protein ensemble sampler with experimental measurements

In this section we propose ExEnDiff, a simple framework that aims to enhance existing diffusion-based unified protein ensemble sampler's to generate physically feasible protein conformation samples that follow a true Boltzmann distribution. Built on the manifold constraint sampling technique [126], an experiment-guided sampling strategy is proposed to alter the intermediate score function to match the experimental measurements.

Consider an experimental measurement that can be written as a function of atomic positions, $\mathbf{y} = A(\mathbf{x}_0)$, where $\mathbf{y} = [y_1, \ldots, y_m] = [A_1(\mathbf{x}_0), \ldots, A_m(\mathbf{x}_0)]$ represents m one-dimensional features. With a prior diffusion-based protein ensemble sampler, our aim is to retrieve and generate conformation ensembles from the conditional probability distribution $p(\mathbf{x}_0|A(\mathbf{x_0}) = y)$. If we take the score-based diffusion model as the prior, then we can use the reverse diffusion from Eq. (3) as the sampler from the conditional distribution as follows:

$$d\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y})]dt + g(t)d\bar{\mathbf{w}}, \tag{6}$$

where $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y})$ can be decomposed using the Bayes rule:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t, t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t). \tag{7}$$

By utilizing a trained score model $\mathbf{s}_\theta$ to approximate the score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ and computing the likelihood gradient $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)$, we obtain an expression for the posterior likelihood $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y})$. However, deriving a closed-form solution for $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)$ is challenging due to the implicit dependency between $\mathbf{y}$ and $\mathbf{x}_t$. We next outline an approach to estimate $p_t(\mathbf{y}|\mathbf{x}_t)$.

Recognizing that $\mathbf{y}$ and $\mathbf{x}_t$ are conditionally independent given $\mathbf{x}_0$, we express the conditional probability as:

$$p(\mathbf{y}|\mathbf{x}_t) = \int p(\mathbf{y}|\mathbf{x}_0, \mathbf{x}_t)p(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0 = \int p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0. \tag{8}$$

This formulation allows us to express $p(\mathbf{y}|\mathbf{x}_t)$ as an expectation:

$$p(\mathbf{y}|\mathbf{x}_t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0|\mathbf{x}_t)}[p(\mathbf{y}|\mathbf{x}_0)]. \tag{9}$$

To approximate this expectation, we assume:

$$p(\mathbf{y}|\mathbf{x}_t) \approx p(\mathbf{y}|\hat{\mathbf{x}}_0), \tag{10}$$

where $\hat{\mathbf{x}}_0$ is the posterior mean estimate given by:

$$\hat{\mathbf{x}}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t + (1 - \bar{\alpha}_t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)), \tag{11}$$

as derived in the context of DDPM [126].

The conditional probability $p(\mathbf{y}|\mathbf{x}_0)$ is theoretically modeled using a Dirac delta function $p(\mathbf{y}|\mathbf{x}_0) = \delta(\mathbf{y} - \mathcal{A}(\mathbf{x}_0))$. In practice, to guarantee numerical stability, we approximate this function with a Gaussian distribution of small variance:

$$U_{\text{cons}} = \frac{\|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0)\|_2^2}{2\sigma^2},$$
$$p(\mathbf{y}|\mathbf{x}_0) = \frac{1}{\sqrt{(2\pi)^n\sigma^{2n}}} \exp(-U_{\text{cons}}), \tag{12}$$

where n represents the dimensionality of $\mathbf{y}$ and σ is the standard deviation that regulates the strength of the constraint.

By leveraging the approximation from the earlier expression, the posterior term can be estimated with a harmonic restraint potential function:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) \approx -\tfrac{1}{2}\mathbf{k}\nabla_{\mathbf{x}_t}\|\mathbf{y} - A(\hat{\mathbf{x}}_0)\|_2^2, \tag{13}$$

where $\mathbf{k} = [k_1, \ldots, k_i]$, with $k_i = 2/\sigma_i^2$ the restraint constant of ith function, and $\hat{\mathbf{x}}_0 = \mathbb{E}[\mathbf{x}_0|\mathbf{x}_t]$ is the posterior mean. Finally, with a pretrained diffusion model $\mathbf{s}_\theta(\mathbf{x}_t, t)$ as the prior, we can approximate $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y})$ with a conditional score function $\mathbf{s}_{\text{rest}}(\mathbf{x}_t, t)$ at the inference time:

$$\mathbf{s}_{\text{rest}}(\mathbf{x}_t, t) = \mathbf{s}_\theta(\mathbf{x}_t, t) - \tfrac{1}{2}\mathbf{k}\nabla_{\mathbf{x}_t}\|\mathbf{y} - A(\hat{\mathbf{x}}_0)\|_2^2,$$
$$d\mathbf{x}_t = [\mathbf{f}(\mathbf{x}_t, t) - g^2(t)\mathbf{s}_{\text{rest}}(\mathbf{x}_t, t)]dt + g(t)d\bar{\mathbf{w}}. \tag{14}$$

The correction term weight k_i is an important hyperparameter in the manifold constraint sampling. A too-low weight will lead to inconsistency with the conditions and an overly high weight will make the sampling path noisy and thus generate unphysical configurations. Following the original manifold constraint sampling technique [126], we set $k_i = k_i'/\|\mathbf{y} - A_i(\hat{\mathbf{x}}_0)\|$. In practice, different operators can require different values of k_i' for optimized performance. We report the empirical values that yield the best performance in Supplemental Material [36], Sec. 8.3.

The experimental measurement can give either the probability distribution $P_{\text{measurement}}(\mathbf{y})$ or the ensemble average $\bar{\mathbf{y}}_{\text{measurement}}$. If the experimental measurements give observable average $\bar{\mathbf{y}}_{\text{measurement}}$, then we approximate the probability distribution of $\mathbf{y}$ with a Gaussian distribution centered on $\bar{\mathbf{y}}_{\text{measurement}}$: $P_{\text{measurement}}(\mathbf{y}) = \mathcal{N}(\mathbf{y}; \bar{\mathbf{y}}_{\text{measurement}}, \sigma^2)$, with σ a manually designed variable variance. In all experiments of this paper, we assume the ground-truth probability distribution $P_{\text{measurement}}(\mathbf{y})$ exactly known. We want to enhance the protein ensemble sampler's capacity by forcing it to generate samples that match with the experimental measurement $P_{\text{sample}}(\mathbf{y}) \approx P_{\text{measurement}}(\mathbf{y})$, where

$$P_{\text{sample}}(\mathbf{y}) = \int P_{\text{sample}}(\mathbf{x}_0)\delta(\mathbf{y} - A(\mathbf{x}_0))d\mathbf{x}_0. \tag{15}$$

This can be achieved by the following procedure:

(i) Sample $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N \sim P_{\text{measurement}}(\mathbf{y})$.

(ii) For each $\mathbf{y}_i$, sample $\mathbf{x}_{0_i}^{(1)}, \ldots, \mathbf{x}_{0_i}^{(M)} \sim p(\mathbf{x}_0|A(\mathbf{x}_0) = \mathbf{y}_i)$.

The first step can be achieved easily, since $\mathbf{y}$ is usually of low dimension. For the more challenging second step, we utilize a state-of-the-art unified protein ensemble sampler as the prior and apply manifold constraint sampling techniques. This two-step sampling procedure ensures we have a information-rich protein ensemble sampler to start with and can greatly enhance the model's capacity to generate protein ensembles that match with experiment measurements at no additional computational cost. In all experiments guided by low-dimensional features, we set $N = 20$, $M = 5$. In the experiments guided by cryo-EM 2D images, we set $N = 100$, $M = 1$. Additionally, in the computational cryo-EM model used in this paper, the rotation matrix is prechosen and fixed, while in a true model, the rotation matrix is drawn from the uniform distribution on SO(3). We include full details of ExEnDiff in Supplemental Material [36], Sec. 1, and choice of restraint constant k_i in Supplemental Material [36], Sec. 8.2. It is important to note that if the restraint constant is mischosen or the pretrained model encounters challenging regions of protein space, then it may generate unphysical protein configurations with bond breaks. To mitigate this, we enforce a restraint potential that maintains the distance between consecutive α carbons at approximately 3.8 Å, effectively preventing bond breaking. Furthermore, no steric clashes have been observed [36].

C. Calculation of sampler score

To compare the relative accuracy of each sampler, we report the sampler scores. The sampler scores for IDP are computed as the following:

$$\begin{aligned}\textbf{Score} = {} & \textbf{JS}_{\text{eed}} + \textbf{JS}_{\text{rg}} + \textbf{JS}_{\text{sasa}} \\ & + \frac{1}{N}\sum_{i=1}^{N} \frac{\left|\textbf{RMSF}_{i_{\text{sampler}}} - \textbf{RMSF}_{i_{\text{True}}}\right|}{\textbf{RMSF}_{i_{\text{True}}}} \\ & + \frac{1}{N}\sum_{i=1}^{N} \left|\%\textbf{helix}_{i_{\text{sampler}}} - \%\textbf{helix}_{i_{\text{True}}}\right|. \end{aligned} \tag{16}$$

The score is a sum of five terms: The first three terms are the JS distances with respect to the end to end distance, radius of gyration, and the solvent accessible surface area. These JS distances are measured to quantify the similarity between the distributions of protein conformation ensembles generated by the sampler and those of the ground truth. These terms provide a global measure of how well the sampler captures the overall structural features of the protein ensemble. The fourth term is the residue-specific deviation in RMSF, which measures the difference in flexibility at each residue between the sampled and true ensembles. The fifth term captures the discrepancy in helix content per residue, indicating how well the sampler reproduces the secondary-structure elements. These latter two metrics focus on local structural differences, providing a finer resolution of the sampler's accuracy at the residue level. Together, these components offer a comprehensive evaluation of the sampler's performance in capturing both global and local features of protein conformations.

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DATA AVAILABILITY

The benchmark protein datasets, excluding RS-peptide, were obtained from previously published studies [71,72,73,44]. The RS-peptide trajectory data are openly available on Zenodo [46]. All scripts and code used in this study are available on GitHub [131]. The data that support the findings of this article are openly available [46,131]; embargo periods may apply.

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